Sorting and climate: a robust differentiation signal, but a leverage-driven sorting overlap

Draft results summary (Module C, the extrinsic-climate arm). Methods are deferred to Supplementary Materials. Findings are descriptive: associations with climate are genotype–environment correlations, and the sorting itself is licensed as selection only against the forthcoming neutral null. The climate-axis identities (which variables load on PC1 vs PC2) are pending and will be added (Module C4).

Results

Design and a controlled ancestry confound. We associated per-population allele frequencies with two per-population climate principal components (PC1, PC2) using BayPass, across eight configurations (two population sets × two PCs × two covariance settings). A climate-association *outlier cluster* is an LD cluster (eMLG, ≥ 5 loci) with ≥ 10 member loci individually significant at BF(dB) ≥ 20; sorting is the eMLG-consensus call for every such cluster. Because sorting is, by definition, ancestry fixation, a climate axis that is itself correlated with genome-wide ancestry could co-localise with sorting for structural rather than ecological reasons. PC1 is only weakly ancestry-correlated (Spearman $\rho = -0.23$), but PC2 is moderately so ($\rho = +0.57$), driven by a single population, Åland (dropping it lowers the correlation to +0.38; dropping Sielva instead *raises* it; Fig. 1). This shared-ancestry structure is conditioned out by BayPass's covariance matrix (Ω), estimated on the LD-pruned set, and by the Åland-excluded population set; we therefore treat Åland-excluded, with-Ω as the structure-robust primary and read the other configurations as uncontrolled comparators.

Diagnostic-index enrichment is robust for PC1, fragile for PC2. Climate-outlier clusters are enriched for ancestry-diagnostic (high-DI) loci (Fig. 2a). For PC1 this enrichment is robust across configurations (size-adjusted odds ratio 1.8–3.1, all significant). For PC2 it is not: the odds ratio swings from 10.5 to 0.5 between covariance settings and, under the structure-robust primary, reverses to depletion. Enrichment for static ancestry-informativeness is therefore a genuine PC1 (climate) signal, whereas the PC2 enrichment is largely structure-driven.

The sorting–climate overlap is driven by Åland and does not survive the controls. Testing whether outlier clusters are more likely to be directionally sorted (size-adjusted), the only significant overlap is PC2 with Åland included (odds ratio 6.8–16; Fig. 2b). Excluding Åland collapses the PC2 outlier set from ∼15–22 clusters to two, leaving the overlap untestable; PC1 straddles the null throughout. A member-level analysis (fraction of a cluster's member loci that sort) gives no support in either axis (coefficients negative, non-significant), and the sorted primary-outlier clusters are few and *F. polyctena*-leaning (Fig. 2c). The previously reported extrinsic co-localisation of sorting with the PC2 climate axis is thus a leverage artifact of a single population rather than a robust genome-wide signal.

Interpretation and outlook. The robust climate signal in these data is a *static* one—outlier clusters are enriched for ancestry-diagnostic loci along PC1—not a *dynamic* co-localisation of directional sorting with climate. The apparent PC2 sorting overlap does not survive control for the shared-ancestry structure and the Åland leverage point. Two caveats temper the deflation: the strict outlier definition yields few clusters per configuration (2–19), so power is low, especially after population exclusion; and this rests on a single clustering. The extrinsic-climate hypothesis for the *sorting* is therefore, on present evidence, weakly supported relative to the descriptive directional-sorting result itself. Naming the PC1/PC2 climate axes (Module C4) and the recombination-matched neutral null remain the decisive next steps; discriminating intrinsic incompatibilities is pursued separately through among-population linkage disequilibrium between unlinked diagnostic units.

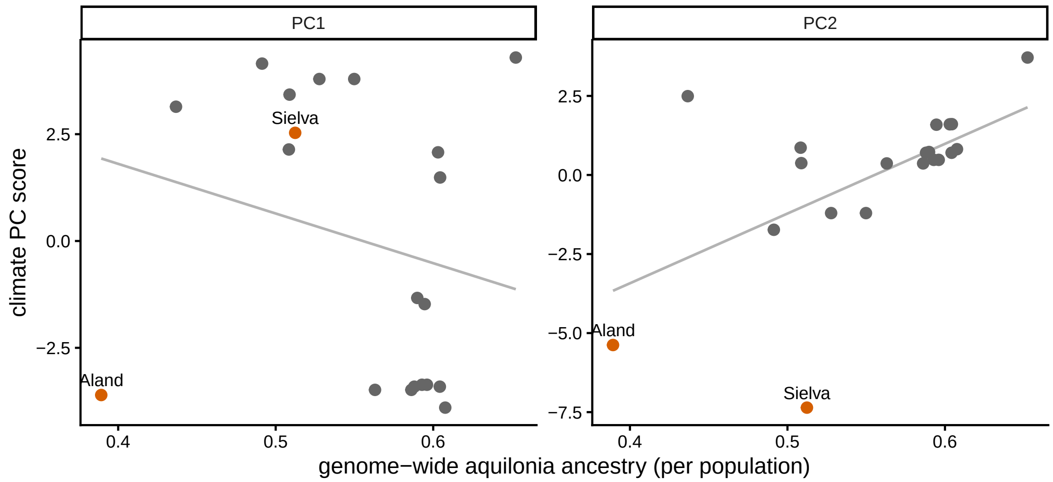

Figure 1: **The climate covariates versus genome-wide ancestry.** Per-population climate PC scores against genome-wide *F. aquilonia* ancestry; Åland and Sielva (orange) highlighted. PC1 is weakly ancestry-correlated; PC2 is moderately so, driven by Åland, motivating the Ω and Åland-excluded controls.

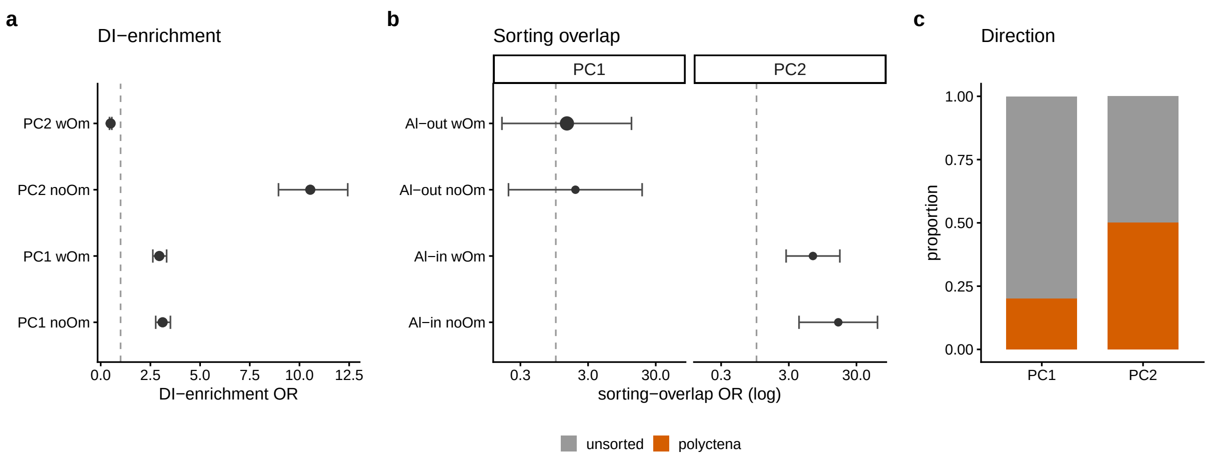

Figure 2: **Sorting and climate outliers.** (**a**) Size-adjusted DI-enrichment of climate-outlier clusters (Åland-excluded set): robust for PC1, fragile/structure-dependent for PC2. (**b**) Size-adjusted odds ratio that an outlier cluster is directionally sorted, by configuration (larger point = structure-robust primary); the significant PC2 overlap appears only with Åland included (Al-in) and is untestable once Åland is excluded. (**c**) Direction of the sorted primary-outlier clusters (few, *F. polyctena*-leaning).